

Keep-beside-you reference sheet

Symbol / word	Meaning and units
System / closed / isolated	Chosen matter / no matter exchange / no matter or energy exchange
U	Internal energy of sample, J; $\Delta U = q + \text{work on sample}$
H	Enthalpy, $U + pV$, J; not a universal synonym for heat
S; T	Entropy, J/K; absolute temperature, K
F	Helmholtz energy, $U - TS$, J; fixed T,V comparison (some texts write A; here A is a component)
G	Gibbs energy, $H - TS$, J; fixed T,p comparison
n; $g = G/n$	Amount, mol of atoms; molar Gibbs energy, J/mol
h; s	Molar enthalpy J/mol; molar entropy J/(mol K)
f _S , f _L	Nonnegative phase amount fractions (not composition mole fractions), summing to one
Equilibrium	Minimum appropriate potential among allowed states under constraints; not a rate
Model / algorithm / output	Physical expressions and parameters / calculation procedure / numerical result to check
CALPHAD	CALculation of PHase Diagrams; phase models and equilibrium calculations, with model assessment requiring evidence

The potential criteria assume simple bulk conditions and fixed component amounts as in W3. Only the synthetic A model W4 is used for numerical phase comparison. Its 1000 K crossing fixes equal energies, not a unique fraction.

Sources for the condensed explanations: [1] R. Jaramillo, MIT 3.020, Lecture 3, 2021, First Law¹. [2] R. Jaramillo, MIT 3.020, Lecture 5, 2021, Entropy². [3] R. Jaramillo, MIT 3.020, Lecture 6, 2021, Thermodynamic potentials³. No DOI is listed for these lecture notes; wording/examples here are original. [4] The course's synthetic model contract defines our assumptions. [5] SciPy developers, linprog⁴. API roles are also documented by the pycalphad developers: phase evaluation⁵ and equilibrium⁶.

¹https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l03.pdf

²https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l05.pdf

³https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l06.pdf

⁴<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html>

⁵<https://pycalphad.org/docs/latest/examples/LegacyEnergySurface.html>

⁶<https://pycalphad.org/docs/latest/examples/EquilibriumWithOrdering.html>